4 Geometry and trigonometry

	Students should be taught to:	Notes
4.1 Angles, lines and triangles	**See Foundation Tier**	
4.2 Polygons	**See Foundation Tier**	
4.3 Symmetry	**See Foundation Tier**	
4.4 Measures	**See Foundation Tier**	
4.5 Construction	**See Foundation Tier**	
4.6 Circle properties	**A** understand and use the internal and external intersecting chord properties	
	B recognise the term 'cyclic quadrilateral'	
	C understand and use angle properties of the circle including: (i) angle subtended by an arc at the centre of a circle is twice the angle subtended at any point on the remaining part of the circumference (ii) angle subtended at the circumference by a diameter is a right angle (iii) angles in the same segment are equal (iv) the sum of the opposite angles of a cyclic quadrilateral is $180°$ (v) the alternate segment theorem	Formal proof of these theorems is not required
4.7 Geometrical reasoning	**A** provide reasons, using standard geometrical statements, to support numerical values for angles obtained in any geometrical context involving lines, polygons and circles	
4.8 Trigonometry and Pythagoras' theorem	**A** understand and use sine, cosine and tangent of obtuse angles	
	B understand and use angles of elevation and depression	

	Students should be taught to:	Notes
	C understand and use the sine and cosine rules for any triangle	
	D use Pythagoras' theorem in three dimensions	
	E understand and use the formula $\frac{1}{2}ab\sin C$ for the area of a triangle	
	F apply trigonometrical methods to solve problems in three dimensions, including finding the angle between a line and a plane	The angle between two planes will not be required
4.9 Mensuration	**A** find perimeters and areas of sectors of circles	Radian measure is excluded
4.10 3D shapes and volume	**A** find the surface area and volume of a sphere and a right circular cone using relevant formulae	
4.11 Similarity	**A** understand that areas of similar figures are in the ratio of the square of corresponding sides	
	B understand that volumes of similar figures are in the ratio of the cube of corresponding sides	
	C use areas and volumes of similar figures in solving problems	